

VENKATRAMANAN MAHENDRARAJAN

Scientist (ICMR Project)



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Research Metrics Citations: 232, h-index: 6, i10-index: 4



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<https://venkatramananmahendrarajan.github.io>

ABOUT ME

I am a dedicated researcher with expertise in microbiology, molecular biology, bioinformatics and animal handling for preclinical studies. My work spans bacterial membrane vesicles, antimicrobial peptides, gene expression analysis, and computational drug discovery. I have extensive experience in sequencing data analysis for Illumina and Nanopore platforms, as well as transcriptomics-based differential gene expression analysis. Passionate about translational research, I aim to bridge computational and experimental biology for innovative solutions in biotechnology and healthcare.



EDUCATION

Integrated MSc Life Sciences

Central University of Tamil Nadu

2015-2020

CGPA 8.3/10, First class with Distinction



SKILLS

Microbial techniques

Molecular techniques

Animal Handling

Computational - Docking & Dynamics

Sequencing - Nanopore & Illumina

Microscopy - CLSM & SEM

PCR - Conventional & q RT PCR

Flow cytometry

Proteomics - SDS PAGE & Western blot



PROFESSIONAL EXPERIENCE

2024 - Present	ICMR Project Scientist (Doctoral Research Fellow) Vellore Institute of Technology, Vellore
2022 - Present	Doctoral Research Fellow Vellore Institute of Technology, Vellore
2021 - 2022	SERB Junior Research Fellow Central University of Punjab, Punjab



EXAM QUALIFIED

Gradual Aptitude Test for Engineering (GATE) - Life Sciences, 2020, India.
Percentile - 78.6



TEACHING EXPERIENCE

2022 - 2023	Teaching cum Research Assistant Vellore Institute of Technology
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PUBLICATIONS

- Venkatramanan M**, Sankar Ganesh P, Senthil R, Akshay J, Veera Ravi A, Langeswaran K, et al. Inhibition of Quorum Sensing and Biofilm Formation in *Chromobacterium violaceum* by Fruit Extracts of *Passiflora edulis*. ACS Omega 2020;5:25605-16. <https://doi.org/10.1021/acsomega.0c02483>. IF: 5.2
- Mahendrarajan V**, Bari VK. A critical role of farnesol in the modulation of Amphotericin B and Aureobasidin A antifungal drug susceptibility. Mycology 2022;13:305-17. <https://doi.org/10.1080/21501203.2022.2138599>. IF: 5.7
- Venkatramanan M**, Nalini E. Regulation of virulence in *Chromobacterium violaceum* and strategies to combat it. Front Microbiol 2024;15:1-13. <https://doi.org/10.3389/fmicb.2024.1303595>. IF: 5.8
- Mahendrarajan V**, Lazarus H, Easwaran N. Quorum quenching mediated biofilm impediment in *Chromobacterium violaceum* and *Staphylococcus aureus* by leaf extracts of *Delonix elata*. Heliyon 2024;10:e31898. <https://doi.org/10.1016/j.heliyon.2024.e31898>. IF: 4
- Mahendrarajan V**, Easwaran N. Quorum quenching effects of linoleic and stearic acids on outer membrane vesicle-mediated virulence in *Chromobacterium violaceum*. Biofouling 2024;1-11. <https://doi.org/10.1080/08927014.2024.2446930>. IF: 2.6
- Mahendrarajan V**, Nalini E. In silico analysis of bacteriocins from *Lactobacillus acidophilus* membrane vesicles against *Streptococcus mutans* GtfB protein. J Biomol Struct Dyn 2025;0:1-11. <https://doi.org/10.1080/07391102.2025.2460743>. IF: 2.7



LANGUAGE

- Tamil (Native)
- English (Proficient)
- Hindi (Intermediate)



PROFESSIONAL MEMBERSHIP

2024 - Present	Contributing Membership American Society of Microbiology
2025 - Present	EXO-RISING Membership Indian Extracellular Vesicles Society



SCIENTIFIC PRESENTATIONS

2022	Poster Presentation IMMUNOCON- Annual Conference of the Indian Immunology Society, PIGMER, Chandigarh.
2023	Poster Presentation International Conference on Biotechnology for Environment & Health (ICBEH), VIT, Vellore.
2024	Poster Presentation International Conference on Contemporary Antimicrobial Research (ICCAR), NEHU, Shillong.



RESEARCH TRAININGS

2024	Extracellular Vesicles: Isolation, Characterization & Applications, AMRF, Madurai
2025	Training on the state of art advanced multi omics techniques in Molecular Oncology, ICMR, Govt. of India, Cancer Institute (WIA), Chennai.



ONGOING RESEARCH WORK

Parkinson's Disease Multi-Omics Integration

Conducted integrative multi-omics analysis combining brain transcriptomics and gut microbiome metagenomics datasets from control and Parkinson's disease cohorts to investigate the gut-brain axis. Performed differential gene expression analysis using DESeq2 and carried out comprehensive microbiome profiling using HUMAnN and MetaPhlAn. Mapped host-microbiome interactions to identify functionally relevant associations contributing to neuroinflammation. Applied machine learning approaches, including Random Forest-based feature selection, to identify key predictive signatures and potential biomarkers associated with Parkinson's disease pathology

[7] **Mahendrarajan, V.**, Kodiveri Muthukaliannan, G., & Easwaran, N. (2025). Quorum quenching mediated virulence factor impairment by Pandanus tectorius leaf extract against Chromobacterium violaceum . Natural Product Research, 1–8. <https://doi.org/10.1080/14786419.2025.2491837>. IF: 2

[8] **Venkatramanan, M** & Easwaran, N. (2025). Isolation, Probiotic Characterization and Genomic Analysis of Enterococcus durans VIT3 from Edible Curd. Microbial Pathogenesis. 107649. 10.1016/j.micpath.2025.107649. IF: 3.9

[9] **Mahendrarajan V**, Lazarus HPS, Muthukaliannan GK, Varghese S, Easwaran N. Membrane vesicles from Red Complex bacteria: key players in oral pathogenesis, immune disruption, systemic diseases, and therapeutic insights. Front Oral Health. 2025 Jul 28;6:1607931. doi: 10.3389/froh.2025.1607931. IF: 3.4

[10] **Mahendrarajan, V.** and Easwaran, N. (2026) 'Integrated proteomics, molecular dynamics, and in vitro characterization of antimicrobial peptide from Lactobacillus acidophilus vesicles against Streptococcus mutans', RSC Adv., 16(7), 5743–5757. <https://doi.org/10.1039/D5RA09511E>. IF: 6.1

[11] Lazarus, H. P. S., **Mahendrarajan, V.**, & Easwaran, N. (2026). Genomic insights into Exiguobacterium indicum VIT-2023: a dual-function PGPR with potential for plant health protection and plastic biodegradation. Environmental Pollutants and Bioavailability, 38(1). <https://doi.org/10.1080/26395940.2026.2636356>. IF: 4

[12] **V. Mahendrarajan**, N. Easwaran, Biocompatible Membrane Vesicles from Lactobacillus acidophilus MTCC 10307 Exhibit Potent Anti-Inflammatory Activity, BioRxiv. (2026). <https://doi.org/10.64898/2026.04.01.715785>.

[13] **Mahendrarajan V**, Easwaran N. Lactobacillus acidophilus membrane vesicles demonstrate antimicrobial and antibiofilm activity against Streptococcus mutans – an in vitro and in vivo study. Artif Cells Nanomed Biotechnol. 2026 Dec;54(1):17–35. doi: 10.1080/21691401.2026.2662204. Epub 2026 Apr 29. PMID: 42052939. IF: 5.6

[14] **Mahendrarajan, V.**, Sankaranarayanan, G., Muthukaliannan, G.K. et al. (2026). Triphala Modulates the Membrane Vesicle Transcriptome of Enterococcus durans VIT3 to Influence Antibiotic Response and Probiotic Functions. Mol Biotechnol. <https://doi.org/10.1007/s12033-026-01578-x>. IF: 3.2



PATENT PUBLISHED

[1] Easwaran, N., **Mahendrarajan, V.**, and Lazarus, H.P.S., A process for extraction and purification of outer membrane vesicles from complex biofilm architectures of Chromobacterium violaceum. Application No.202541131837 A

[2] Easwaran, N. and **Mahendrarajan, V.**, A synthetic anticacterial peptide VMNE244 derived from Lactobacillus acidophilus membrane vesicles and its use against Streptococcus mutans



REFERENCES

Dr. Nalini. E – Vellore Institute of Technology, Assistant professor
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Dr. Gothandam. K. M – Vellore Institute of Technology, Professor
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